

WAVES – PROPERTIES OF WAVES

KEY VOCABULARY

Wavelength (λ): distance from one wave crest to the next, in metres.

Frequency (f): number of waves per second, measured in hertz (Hz).

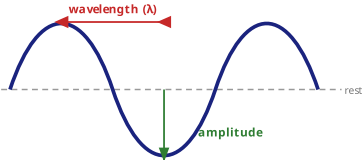
Amplitude: maximum displacement from the rest position.

Transverse: vibrations are at 90° to the direction of travel.

COMMON MISCONCEPTIONS

- ~~Waves transfer matter from place to place.~~
- ✓ Waves transfer energy, not matter.
- ~~Amplitude is the same as wavelength.~~
- ✓ Amplitude is height; wavelength is length.
- ~~Sound is a transverse wave.~~
- ✓ Sound is a longitudinal wave.

PARTS OF A TRANSVERSE WAVE



Wave speed: $v = f \times \lambda$ (speed = frequency $\times$ wavelength).

1 LABEL THE WAVE

On the diagram above, name the parts marked **wavelength** and **amplitude**. [2 marks]

2 WAVE SPEED

A wave has frequency 50 Hz and wavelength 0.4 m. **Calculate** its speed. [2 marks]

Answer: _____ m/s

3 TRANSVERSE VS LONGITUDINAL

Describe the difference between transverse and longitudinal waves. [2 marks]

4 FREQUENCY & PERIOD

A wave has a period of 0.02 s. **Calculate** its frequency ($f = 1 \div T$). [2 marks]

Answer: _____ Hz

5 ENERGY TRANSFER

Explain what a wave transfers when a cork bobs on water but does not move along. [2 marks]

6 EM SPECTRUM

Complete the sentence using the word bank. [2 marks]

All electromagnetic waves are _____ and travel at _____ m/s in a vacuum.

transverse

longitudinal

300 000 000

340